

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
MID SEMESTER EXAMINATION July-Dec 2023

SUBJECT: INTRODUCTION TO COMPUTING

DURATION: 2 Hrs

SUBJECT CODE: CS15101

FULL MARKS: 30

SECTION: ECE I & II

SEMESTER: Ist

Attempt all questions.

		Marks	CO	BL
Q1.	Explain the components of computer system in brief with suitable block diagram. Compare the different types of memory used in computer system in terms of speed, size, and cost. OR What is the purpose of header file in C program? Is the use of header file necessary? Explain the rough template to write a C code.	[5]	CO1 & CO2	understand
Q2.	Differentiate between the following with suitable examples. a) Implicit vs Explicit type conversion. b) Pre increment/decrement vs Post increment/decrement operator. c) Lvalue vs Rvalue d) Associativity vs precedence of operator e) Pre-test vs Post-test loop	[2*5 =10]	CO2	understand
Q3	Explain constant, variable and keywords. What is the importance of data type in C? Mention the various data types with the amount of memory, format specifier and range it occupies in the system.	[5]	CO2	understand
Q4.	Write a program in C to find whether a character entered by a user is vowel or not using if-else and switch-case control sequence both. Which code sequence is better and why?	[4]	CO2	understand
Q5.	Identify and correct the errors, then write the output of following codes. a) <code>#include<stdio.h></code> <code>int main()</code> <code>{</code> <code>int n=345, s=0, r;</code> <code>printf("\n enter the number");</code> <code>scanf("%d",n);</code> <code>for(;n>0;n/=10)</code> <code>{</code> <code>r=n%10;</code> <code>s=s+r;</code> <code>}</code> <code>printf("\n%d",s);</code> <code>return 0;</code> <code>}</code> b) <code>#include<stdio.h></code> <code>int main()</code> <code>{</code> <code>int i;</code> <code>for(i=5; ++i; i+=3)</code> <code>print("%d",i);</code> <code>return 0;</code> <code>}</code> c) <code>#include<stdio.h></code> <code>int main()</code> <code>{</code> <code>int i=3;</code> <code>for(i--; i<7; i=7)</code> <code>printf("%f",i++);</code> <code>return 0;</code> <code>}</code>	[2*3 =6]		Understand and apply

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Electronics and Communication Engineering
MID SEMESTER EXAMINATION (October-2023)

B.Tech: ECE (Semester-I)

Course Name: Electronics Workshop

Maximum Time: 2 hours

Date: 17-10-2023

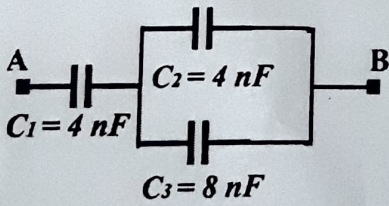
Roll No.:

Course Code: EC15103

Max. Marks: 30 marks

Instruction:

1. **Attempt all questions.**
2. Assume any suitable data, if necessary. *(Any other Instruction need to provide by the concerned faculty)*
3. The Marks, CO (Course Outcome), and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

Q1.	Explain the working of cathode ray oscilloscope (CRO), with proper internal block diagram.	(CO-1) 6 marks
Q2.	What is soldering and de-soldering?, further, discuss the component required for soldering.	(CO-2) 4 marks
Q3.	(a) What is difference between active and passive component, with some examples. (b) Describe the LCR-Q-meter, and find the series resonance frequency.	(CO-1) 2+4 marks
Q4.	Explain the voltmeter circuit, with proper diagram.	(CO-1) 4 marks
Q5.	(a) Design a Pi-type symmetrical attenuator to determine its resistance (R_1 , R_2 , and R_3) values, if voltage gain (A_{dB}) is 20 dB, and output resistance (R_o) is 100 Ω . (b) Determine the equivalent capacitance?  $C_1 = 4 \text{ nF}$ $C_2 = 4 \text{ nF}$ $C_3 = 8 \text{ nF}$	(CO-1) 4+1 marks
Q6.	Design the light operated relay circuit.	(CO-1, CO-3) 5 marks

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

DEPARTMENT OF MATHEMATICS

MID-SEMESTER EXAMINATION - OCTOBER 2023

COURSE: ENGINEERING MATHEMATICS

CODE: MA15102/ MA19101

TIME: 2 HOUR

MAXIMUM MARKS: $5 \times 6 = 30$

ANSWER ALL QUESTIONS

1. Reduce the following matrix

$$A = \begin{pmatrix} 1 & 4 & 3 & -2 & 1 \\ -2 & -3 & -1 & 4 & 3 \\ -1 & 6 & 7 & 2 & 9 \\ -3 & 3 & 6 & 6 & 12 \end{pmatrix}$$

into normal form and hence find its rank.

2. Obtain the value of K for the following system of linear equations

$$\begin{aligned} x + 2y - 4z - w &= 7 \\ -x + 10y + Kz - 4w &= -16 \\ 2x + 5y - 5z + w &= 2 \\ 4x + 11y - 7z + Kw &= 7 \end{aligned}$$

for which the system has (i) unique solution, (ii) infinitely many solution, and (iii) no solution.

3. Determine whether the set $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : 2a_1 - 7a_2 + a_3 = 0\}$ is a subspace of $\mathbb{R}^3$ under the operations of addition and scalar multiplication defined on $\mathbb{R}^3$. Justify your answers. If W is a subspace, then determine the basis and dimension of W .
4. Prove that the transformation $T : P_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ defined by $T(a_0 + a_1x + a_2x^2) = (a_0, a_1, a_3)$ is linear. Let $\beta = \{1, 1+x, 1+x+x^2\}$ and $\gamma = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$ are the ordered bases for $P_2(\mathbb{R})$ and $\mathbb{R}^3$, respectively. Compute $[T]_{\gamma}^{\beta}$.
5. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

6. Let T be a linear operator on the vector space $\mathbb{R}^3$ defined by

$$T(a, b, c) = (3a + 2b - 2c, -4a - 3b + 2c, -c)$$

and $\beta = \{(0, 1, 1), (1, -1, 0), (1, 0, 2)\}$ be the ordered basis. Compute $[T]_{\beta}$, and determine whether β is consisting of eigenvectors of T .

***** ❖ *****

Sub: ENGINEERING CHEMISTRY

Code: CH15101

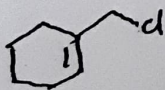
Course: B.Tech (ECE-2)
2nd Year

ANSWER ALL QUESTIONS

$$4\frac{1}{2} + 3 = 7\frac{1}{2}$$

Time = 2 hours

F.M. = 30

1. (a) How can you determine percentage of nitrogen, sulphur and ash by ultimate analysis.
- (b) 2.5 g of air dried coal sample was taken in a silica crucible, after heating it in an electric oven at 110°C for one hour, the residue was weighed 2.410 g. The residue was heated in a silica crucible covered with vented lid at temperature 950°C for exactly seven minutes. After cooling the weight of residue was found to be 1.78 g. The residue was ignited at 750°C to a constant weight of 0.246 g. Calculate results of proximate analysis.
 $1 + 1 + 4\frac{1}{2} + 2 + 2 = 7\frac{1}{2}$
2. (a) What do you understand by Specific Conductance, Equivalent Conductance & Kohlrausch's law.
- (b) The molar conductivity of 0.05 M solution of $MgCl_2$ is $194.2 \text{ S cm}^2 \text{ mol}^{-1}$ at 25°C. A cell with electrodes that are 1.50 cm^2 in surface area and 0.50 cm apart is filled with 0.05 M $MgCl_2$ solution. How much current will flow when the potential difference between the two electrodes is 5.0 volt?
- (c) The molar conductivity at infinite dilution of $AgNO_3$, $NaCl$ and $NaNO_3$ is 116.5, 110.3 and $105.2 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The conductivity of $AgCl$ in water is $2.4 \times 10^{-6} \text{ S cm}^{-1}$ and of water used in experiment is $3.2 \times 10^{-6} \text{ S cm}^{-1}$. Find out the solubility of $AgCl$ in g/dm^3 .
 $1\frac{1}{2} + 2 + 4 = 7\frac{1}{2}$
3. (a) Explain Bent's rule & VSEPR theory.
- (b) Write & draw structure & shape of:
 SOF_4 , SF_4 , PCl_3F_2 , XeO_3F_2 .
 $3\frac{1}{2} + 2 + 2 = 7\frac{1}{2}$
4. (a) Explain S_N1 & $E1$ reaction mechanism.
- (b) Write reaction mechanism for the most stable product of the following:
- 
 $\xrightarrow[\text{Low temp.}]{\text{aq. KOH}}$?
- (c) ~~When 1-bromo-2,2-dimethylcyclopentane is heated at high temperature in ethanol, which product is produced? Also write its reaction mechanism.~~



Department of Electronics and Communication Engineering, NIT Patna
Mid Semester Examination
Elements of Electronics Engineering (EC15102)
B.Tech, (EC), Semester –I

Max Marks: 30

Time: 2:00 Hrs.

Note: All questions carry equal marks. Any missing data may be assumed suitably.

- 1 (a) Explain the working of clipper with suitable example.
 (b) A germanium diode carries a current of 8mA when it is forward biased with 0.3V at 27°C. (i) Find reverse saturation current. (ii) Find the forward bias voltage required to get a forward current of 100 mA at the same 35°C temperature.
- 2 (a) Explain the working full-wave bridge rectifier and also derived the average and RMS values of the full wave rectifier. Draw the output waveform for a sinusoidal input waveform applied at full-wave rectifier.
 (b) Obtain the output voltage V_o and current through diodes D_1 , D_2 , and $0.33k\Omega$ in Fig. 1.

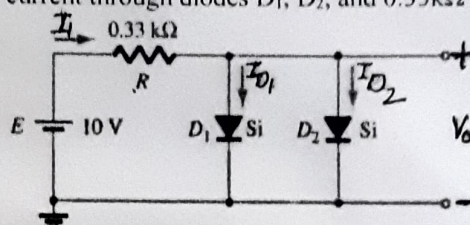


Fig. 1.

- 3 Find the output and draw the waveform of the circuit shown in Fig 2 for the given sinusoidal input. Assume a diode drop of 0.7 V.

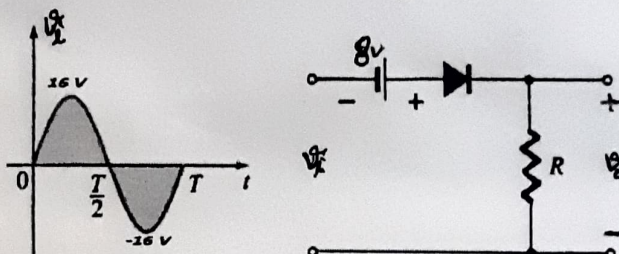


Fig. 2.

- 4 Sketch the output waveform for one complete cycle for the given input & circuit in Fig. 3. Assume the diode is ideal diode.

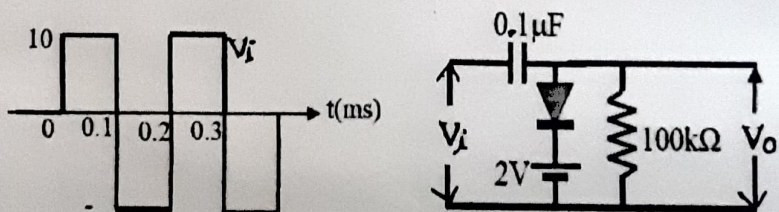


Fig. 3

- 5 (a) Explain in detail about Avalanche breakdown and Zener breakdown phenomena in Diodes.
 (b) Determine current flowing through diode D_2 in the below Fig. 4. Assume the both diodes are Si diodes.

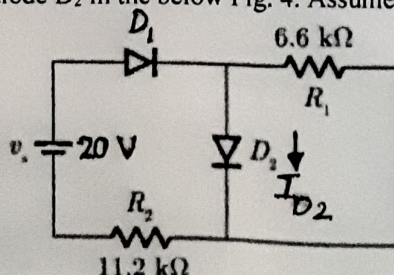


Fig. 4.